

# Does Central Obesity Add Information Beyond BMI for Systolic Blood Pressure, and Does This Relationship Differ by Sex?

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## ABSTRACT

Body mass index summarizes overall body size but does not describe fat distribution. Central adiposity may capture a distinct aspect of body composition that is more closely aligned with cardiovascular risk. This memo evaluates whether a measure of central obesity, the waist-to-hip ratio (WHR), is associated with systolic blood pressure (SBP) after accounting for body mass index. The analysis also assesses whether the waist-to-hip ratio relationship with systolic blood pressure differs between females and males and whether this relationship is linear.

## 1 INTRODUCTION

This report tests whether a measure of fat distribution improves the description of systolic blood pressure (SBP) beyond overall body size. The central question is whether SBP varies systematically with waist-to-hip ratio (WHR) after conditioning on body mass index (BMI), age, sex, and race/ethnicity. Evidence for added information is assessed by comparing the fit of models with BMI alone to models that additionally include WHR, with particular attention to whether the conditional WHR–SBP relationship differs by sex. Because linear specifications can obscure curvature and threshold behavior, spline-based models are used to assess whether the conditional mean of SBP varies nonlinearly with WHR and whether any sex differences occur through the shape of this relationship rather than through a constant slope shift.

## 2 DATA AND VARIABLES

The analysis uses adult participants (age  $\geq 18$  years) with blood pressure and anthropometric measurements available. Systolic blood pressure (SBP) is the primary outcome and is constructed as the within-person mean of up to four SBP readings from the examination component, averaging available replicates to reduce measurement noise. The primary exposure is waist-to-hip ratio (WHR), defined as waist circumference divided by hip circumference. Body mass index (BMI;  $\text{kg/m}^2$ ) is included as a measure of overall body size so that the WHR–SBP relationship can be interpreted as an association with central adiposity after accounting for overall size.

Covariates include age (years), sex (female/male), and race/ethnicity (modeled as categories). These covariates are included to adjust for systematic differences in SBP and body composition across demographic groups. In addition, some model specifications consider the family income-to-poverty ratio (PIR) as a socioeconomic adjustment variable, but PIR is not treated as a primary scientific target; therefore, PIR is used only as a sensitivity adjustment rather than as a centerpiece of the interpretation.

The analytic dataset is defined using complete-case inclusion for the variables required by each model specification, with mild range checks on SBP and key anthropometrics (BMI, waist, hip, and WHR) to limit the influence of implausible values while retaining broad coverage. The resulting sample size is  $n = 4979$  in specifications that require PIR. In the PIR-complete sample, SBP has mean  $\approx 125.9$  mmHg ( $SD \approx 19.7$ ), and participants have mean age  $\approx 49.6$  years ( $SD \approx 18.4$ ), mean BMI  $\approx 29.5$   $\text{kg/m}^2$  ( $SD \approx 7.2$ ), and mean WHR  $\approx 0.94$  ( $SD \approx 0.08$ ), providing scale context for effect-size interpretation. As an initial descriptive comparison that does not rely on parametric adjustment, I summarize mean SBP across WHR tertiles within broad BMI categories. This stratification provides context for the regression analysis by showing whether SBP varies with central adiposity when overall body size is held approximately constant (Figure 1).

### 3 METHODS

The analysis targets the conditional mean relationship between systolic blood pressure (SBP) and central adiposity, measured by waist-to-hip ratio (WHR), while accounting for overall body size and demographic structure. SBP is treated as a continuous outcome, and all findings are interpreted as associations rather than causal effects. Model comparisons are summarized using adjusted  $R^2$  and the Akaike Information Criterion (AIC), where lower AIC indicates better expected out-of-sample fit after penalizing model complexity.

#### 3.1 Linear benchmark and effect modification

A linear regression benchmark is used to assess whether WHR is associated with SBP beyond BMI and whether this association differs by sex. The specification includes BMI, age, race/ethnicity indicators, sex, WHR, and a WHR-by-sex interaction. Heteroskedasticity-robust standard errors are used to reduce sensitivity to non-constant conditional variance.

#### 3.2 Spline-based models for nonlinearity and sex-specific shape

Because anthropometric associations with SBP may be nonlinear, regression spline terms are introduced for WHR (and for age when indicated), yielding a GAM-style specification within the linear modeling framework. To allow sex-specific nonlinear structure, the spline basis for WHR is interacted with sex, permitting distinct smooth functions of WHR for females and males.

Evidence for sex-specific shape differences is assessed using a joint Wald test of the WHR-by-sex spline coefficients. To interpret fitted patterns on the observed scale, I report predicted SBP as a function of WHR with pointwise 95% confidence intervals, evaluated at a reference profile (median age and BMI; modal race/ethnicity). The corresponding prediction curves are presented in the Results section.

#### 3.3 Dimension reduction for high-dimensional laboratory adjustment

NHANES includes a large set of laboratory biomarkers that may capture physiological variation correlated with both adiposity and SBP. Rather than manually selecting biomarkers, a dimension-reduction step is used to summarize the laboratory panel. Laboratory measures with sufficient completeness are standardized, missing values are imputed using within-variable medians, and principal components analysis (PCA) is applied to the standardized laboratory matrix. The first  $k$  principal components are then included as additional adjustment variables in both the linear benchmark and spline-based specifications. This stage evaluates whether the WHR-by-sex patterns and evidence for nonlinear effect modification persist after controlling for broad laboratory-derived physiological variation.

## 4 RESULTS

#### 4.1 Sex-specific, nonlinear WHR–SBP association

In the primary analytic sample, the linear benchmark model with a WHR-by-sex interaction provides a baseline description of the WHR–SBP association. Allowing the WHR association to vary nonlinearly by sex improves fit relative to the linear interaction benchmark, as reflected by a lower AIC and higher adjusted  $R^2$  in the spline-based specification. A joint Wald test of the WHR-by-sex spline coefficients rejects the null hypothesis of a common WHR shape across sexes, providing evidence that the conditional mean pattern of SBP across WHR differs for females and males. The fitted sex-stratified prediction curves highlight that the heterogeneity is expressed through differences in shape rather than only a constant slope shift (Figure 2). Quantitatively, relative to the linear interaction benchmark, allowing a sex-specific WHR spline reduced AIC from 42200.3 to 42180.6 and increased adjusted  $R^2$  from 0.279 to 0.284; the joint Wald test for the WHR-by-sex spline terms strongly rejected a common WHR shape ( $p = 1.3 \times 10^{-8}$ ).

#### 4.2 Robustness to dimension-reduced laboratory adjustment

After adjusting for high-dimensional laboratory measures using principal components, overall fit improves while the sex-specific WHR pattern persists. The PCA-adjusted spline model yields a lower AIC and higher adjusted  $R^2$  than its non-DR counterpart, and the WHR-by-sex spline terms remain jointly significant. The predicted SBP curves from the PCA-adjusted model retain the

same qualitative sex-specific structure observed without laboratory adjustment, indicating that the main WHR heterogeneity is not eliminated by broad adjustment for laboratory-derived physiological variation (Figure 3).

### 4.3 Sufficient-dimension-reduction visualization (SIR scores).

To visualize multivariate anthropometric structure related to SBP without committing to a high-dimensional parametric adjustment, I construct low-dimensional sufficient-dimension-reduction scores using sliced inverse regression (SIR). SIR estimates a small number of linear combinations of anthropometric predictors that retain information about the conditional mean of SBP, under standard linearity and constant-variance conditions for the predictors. SIR therefore provides a complementary, model-light view of multivariate anthropometric structure related to SBP.

These scores are used only for visualization: (i) scatterplots linking SIR score 1 to key anthropometrics with sex-specific LOWESS overlays (Figure 4), and (ii) stratified LOWESS curves of centered SBP across score 2 within strata of score 1 (Figure 5).

## 5 DISCUSSION AND FINDINGS

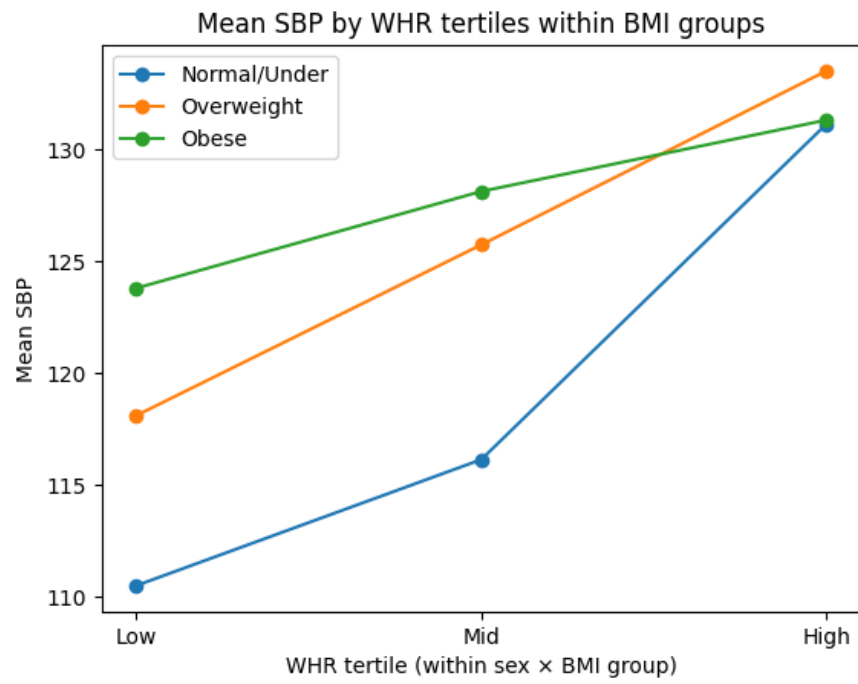
This analysis indicates that WHR is associated with SBP after conditioning on BMI and demographic covariates, supporting the interpretation that fat distribution captures information about SBP beyond overall body size. The stratified descriptive comparison shows that SBP differs across WHR levels even within broad BMI strata, motivating a model-based assessment that separates central adiposity from overall size (Figure 1).

The primary modeling finding is that the WHR–SBP association exhibits both nonlinearity and heterogeneity by sex. Models that allow sex-specific smooth functions of WHR provide a better description of SBP than a linear WHR-by-sex interaction benchmark, and the joint test of sex-specific WHR spline terms supports the conclusion that the fitted WHR pattern differs in shape between females and males (Figure 2). In practical terms, summarizing the WHR–SBP association using only a single linear slope (even with an interaction) can miss meaningful structure in how SBP changes across the observed WHR range.

The sex-specific WHR pattern remains present after adjustment for high-dimensional laboratory biomarkers summarized via principal components (Figure 3). While the laboratory components improve overall model fit, they do not remove the qualitative WHR heterogeneity, suggesting that the observed WHR association is not solely explained by a small set of measured biomarkers and is consistent with WHR capturing a distinct aspect of body composition related to SBP.

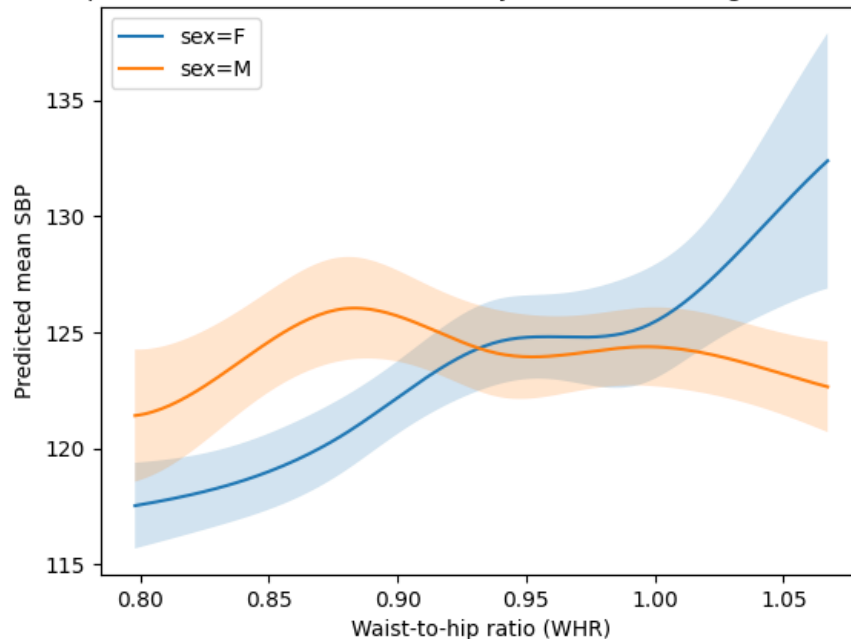
The SIR visualizations provide complementary evidence that multivariate anthropometric variation can be represented in a small number of directions that relate to SBP, and they help interpret how body size measures align with the reduced-dimension structure without treating SIR as a predictive model (Figures 4–5).

All conclusions are based on cross-sectional observational data and are therefore interpreted as associations. SBP is derived from a small number of examination readings; averaging replicates reduces measurement noise but does not eliminate short-term variability. In addition, analyses rely on complete-case inclusion for the variables required by each specification, so the findings pertain to adults with observed SBP and anthropometrics (and, when applicable, laboratory measures). Predicted curves are evaluated at a reference profile (median age and BMI; modal race/ethnicity), which affects the level of the curves but does not determine the presence of sex-specific shape differences across WHR.



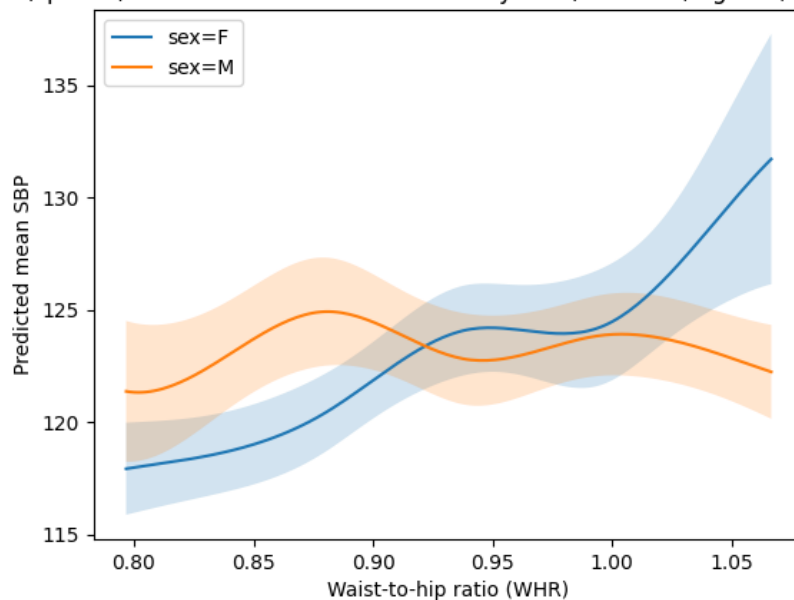
**Figure 1.** Descriptive stratification of systolic blood pressure by central adiposity within broad BMI categories. Points/lines show the mean SBP within WHR tertiles (low/mid/high) computed separately inside each BMI group. This plot provides nonparametric context for the regression models by assessing whether SBP varies with WHR when overall body size (BMI category) is held approximately constant.

GAM (splines): Predicted SBP vs WHR by sex (BMI 28.4, age 51, race=NHW)

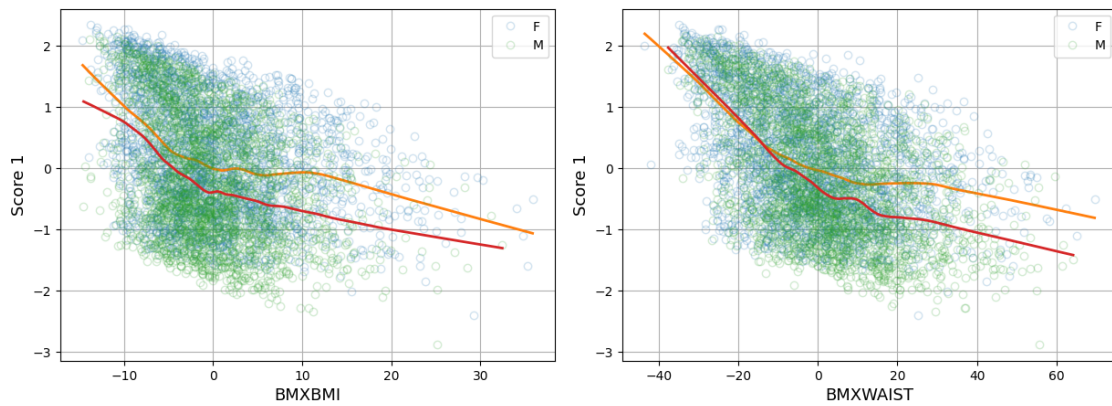


**Figure 2.** Estimated conditional mean SBP as a smooth function of WHR from a spline-based model allowing sex-specific WHR shapes. Curves show fitted values for females and males with pointwise 95% confidence bands. Predictions are evaluated at a reference profile (median age and BMI; modal race/ethnicity) so that differences across WHR primarily reflect the fitted WHR association rather than compositional differences in covariates.

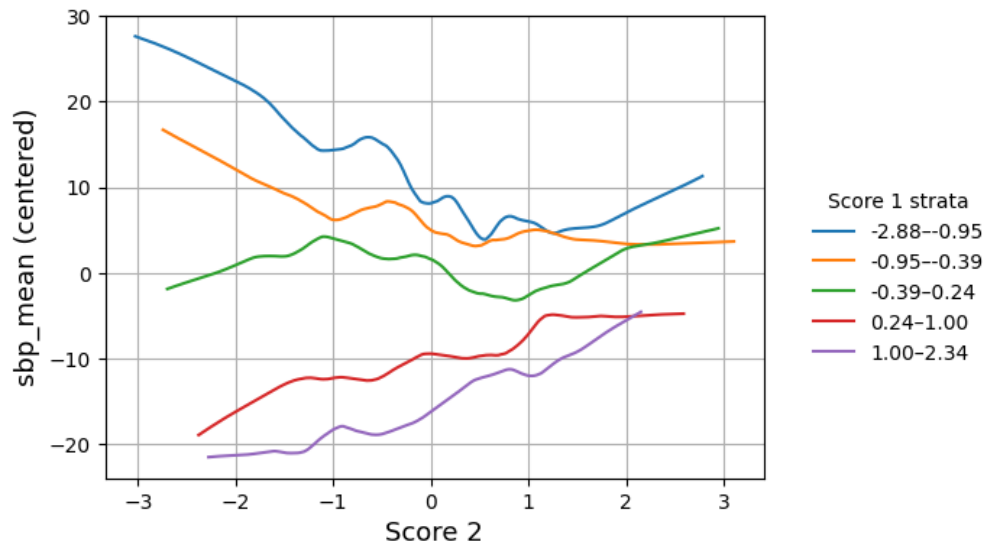
GAM (splines) + DR: Predicted SBP vs WHR by sex (BMI 28.3, age 51, race=NHW)



**Figure 3.** Spline-based predicted SBP–WHR curves by sex after additional adjustment for laboratory biomarkers summarized using principal components. Predictions fix the laboratory principal component scores at 0 (the standardized mean) to isolate the fitted WHR pattern. The persistence of sex-specific WHR shape differences under this adjustment supports robustness of the primary association to broad biomarker-derived physiological variation.



**Figure 4.** Interpretation of the first SIR direction using anthropometric variables. Scatterplots relate SIR score 1 to (left) BMI and (right) waist circumference, with sex-specific LOWESS overlays. These plots help characterize the reduced-dimension score used for visualization by showing how score 1 aligns with common measures of overall size and central adiposity.



**Figure 5.** Stratified LOWESS visualization of SBP across a second SIR score. Curves show centered mean SBP as a function of score 2, drawn separately within strata of score 1 (five groups spanning the empirical distribution of score 1). This plot illustrates how the association between SBP and one reduced-dimension direction varies across levels of the other direction, providing a nonparametric view of multivariate structure related to SBP.